

KHWAJA YUNUS ALI UNIVERSITY

School of Science and Engineering
Department of Computer Science and Engineering
Spring-2023 Final Examination
Program: B.Sc. in CSE (Batch – 14th) 1st Year 2nd Semester
Course Title: Physics - II
Course Code: PHY 1201

PART-A: MCQ

Time: 10 Minutes

Total Marks: 10x1=10

Student's ID:

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Date:

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 Invigilator's Signature:

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(These are single best type of questions. Each of the following questions or statements has five suggested answers or completions. Put tick mark (✓) on the best answer. There is no counter marking.)

Cognitive Knowledge

1. Which is essential for interference?
- A) Two different color lights.
 - B) The movement of the two waves should be opposite.
 - C) The light sources should be far apart.
 - D) The light sources should be coherent.
 - E) The phase difference should have to be $\frac{\pi}{2}$.

Cognitive Analysis

2. When the velocity of the body is equal to light speed, what will be relation between rest mass, m_0 and moving mass, m ?
- A) $m < m_0$
 - B) $m > m_0$
 - C) $m = m_0$
 - D) $m < m_0 < 0$
 - E) $m > m_0 > 0$

Cognitive understanding

3. To happen the photoelectric effect, what condition be essential between incident photon energy (E) and work function (W_0)?
- A) $E < W_0$
 - B) $E \ll W_0$
 - C) $E \leq W_0$
 - D) $E \gg W_0$
 - E) $E \geq W_0$

Cognitive Applying

4. Half-life of a radioactive substance is 10 days. How many days will it take to decrease to 25% of it?
- A) 5
 - B) 10
 - C) 15
 - D) 20
 - E) 25

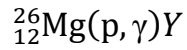
Cognitive Comprehension

5. The slope of the kinetic energy of ejected electron vs frequency of incident photon curve in photoelectric effect is equal to _____.
- A) stopping potential
 - B) threshold frequency
 - C) critical wavelength
 - D) work function
 - E) Planck constant

Please Turn Over (P.T.O.)

Cognitive Applying

6. Identify Y of the following nuclear reaction:



- A) ${}^{27}\text{Al}$
- B) ${}^{23}\text{Na}$
- C) ${}^{26}\text{Mg}$
- D) ${}^{27}\text{P}$
- E) ${}^{27}\text{Mg}$

Cognitive understanding

7. Which is the mirror nucleus of ${}^{15}\text{N}$?

- A) ${}^{14}\text{C}$
- B) ${}^{16}\text{N}$
- C) ${}^{16}\text{O}$
- D) ${}^{14}\text{N}$
- E) ${}^{15}\text{O}$

Cognitive Knowledge

8. Which will be the unit of the decay constant?

- A) m
- B) kg
- C) minutes
- D) sec^{-1}
- E) A

Cognitive Knowledge

9. What is the value of the Compton wavelength (in $\text{\AA}$)?

- A) 0.024
- B) 0.053
- C) 0.24
- D) 0.53
- E) 2.4

Cognitive understanding

10. To reduce the glare of sunglasses, what method of light is used?

- A) Interference
- B) Diffraction
- C) Polarization
- D) Reflection
- E) Superposition

Signature of Examiner

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PART-B: SAQ

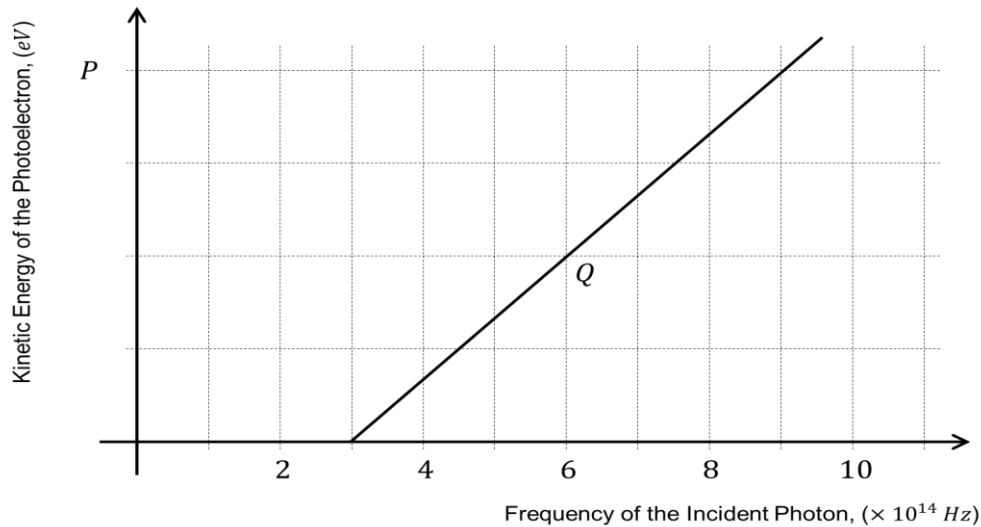
Time: 1 Hour and 50 Minutes

Total Marks: 5x6=30

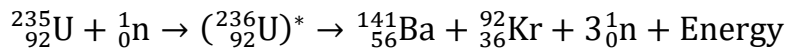
INSTRUCTIONS: Answer question no. 01 (one) and any 04 (four) from the rest. Marks are shown by side.

1. See the stem given below and answer the following questions:

Kinetic Energy of the Photoelectron vs Frequency of the Incident Photon Graph for a Metal



- a) Define photo current. [1]
- b) Write the value of the threshold frequency and find the value of P. [2]
- c) Is it possible to produce the photoelectricity from the metal by using the IR of $1.5 \mu\text{m}$ wavelength? – Justify your answer. [3]
2. See the stem and answer the questions that follows:



Where, $M({}^{235}_{92}\text{U}) = 235.0439 \text{ amu}$; $M({}^{141}_{56}\text{Ba}) = 140.9139 \text{ amu}$;

$M({}^{92}_{36}\text{Kr}) = 91.8973 \text{ amu}$; $M({}^1_0\text{n}) = 1.0087 \text{ amu}$

- a) Give a list of the main components of the nuclear reactor. [1]
- b) Calculate the value of the energy of above stem. [2]
- c) Interpret the process of expansion of the above reaction in the reactor and specify a controlling mechanism for that expansion. [3]
- 3 a) Give some properties of inertial frame of reference. [1]
- b) Explain the theory of relativity is not applicable for the very short velocity of particle compared to the velocity of light. [2]
- c) The total energy of a particle is exactly twice of its rest mass energy. Calculate its speed. [3]

Please Turn Over (P.T.O.)

4. a) What is the activity of the radioactive substances? [1]
b) What will happen when ${}^{238}_{92}\text{U}$ emits an α -particle? [2]
c) Find the activity of 1.00mg of radon, ${}^{222}\text{Rn}$, whose atomic mass is 222 U. [3]
5. a) Draw the binding energy per nucleon curve with proper indications. [1]
b) Explain wave – particle duality. [2]
c) If the velocity of proton is $\frac{1}{20}$ th of the velocity of light, then calculate the de-Broglie wavelength. [3]
6. a) State the Huygens' principle. [1]
b) Sketch the interference, diffraction and polarization phenomena of light. [2]
c) How many orders will be visible (Max. observable angle is 30°) if the wavelength of incident radiation be $5000\text{\AA}$. U . and the number of lines on the grating be 14000 per inch? [3]